

Sorting

Merge Sort

- Merge sort is based divide and conquer strategy.
- Can be implemented as external sorting, when the dataset is so big that it is impossible to load the whole dataset into memory, here sorting is done in chunks.
- Merging is the process of combining two sorted files to make one bigger sorted file.
- Selection is the process of dividing a file into two parts: k smallest elements and $n - k$ largest elements.
- Selection and merging are opposite operations
 - selection splits a list into two lists
 - merging joins two files to make one file

Algorithm – Merge Sort

- 1.Divide:** Divide the list or array recursively into two halves until it can no more be divided.
- 2.Conquer:** Each subarray is sorted individually using the merge sort algorithm.
- 3.Merge:** The sorted subarrays are merged back together in sorted order. The process continues until all elements from both subarrays have been merged.

Index	0	1	2	3	4	5	6
Values	9	4	7	6	3	1	5

1

0	1	2	3
9	4	7	6

4	5	6
3	1	5

2

0	1
9	4

2	3
7	6

12

4	5
3	1

6
5

3

7

13

17

0
9

1
4

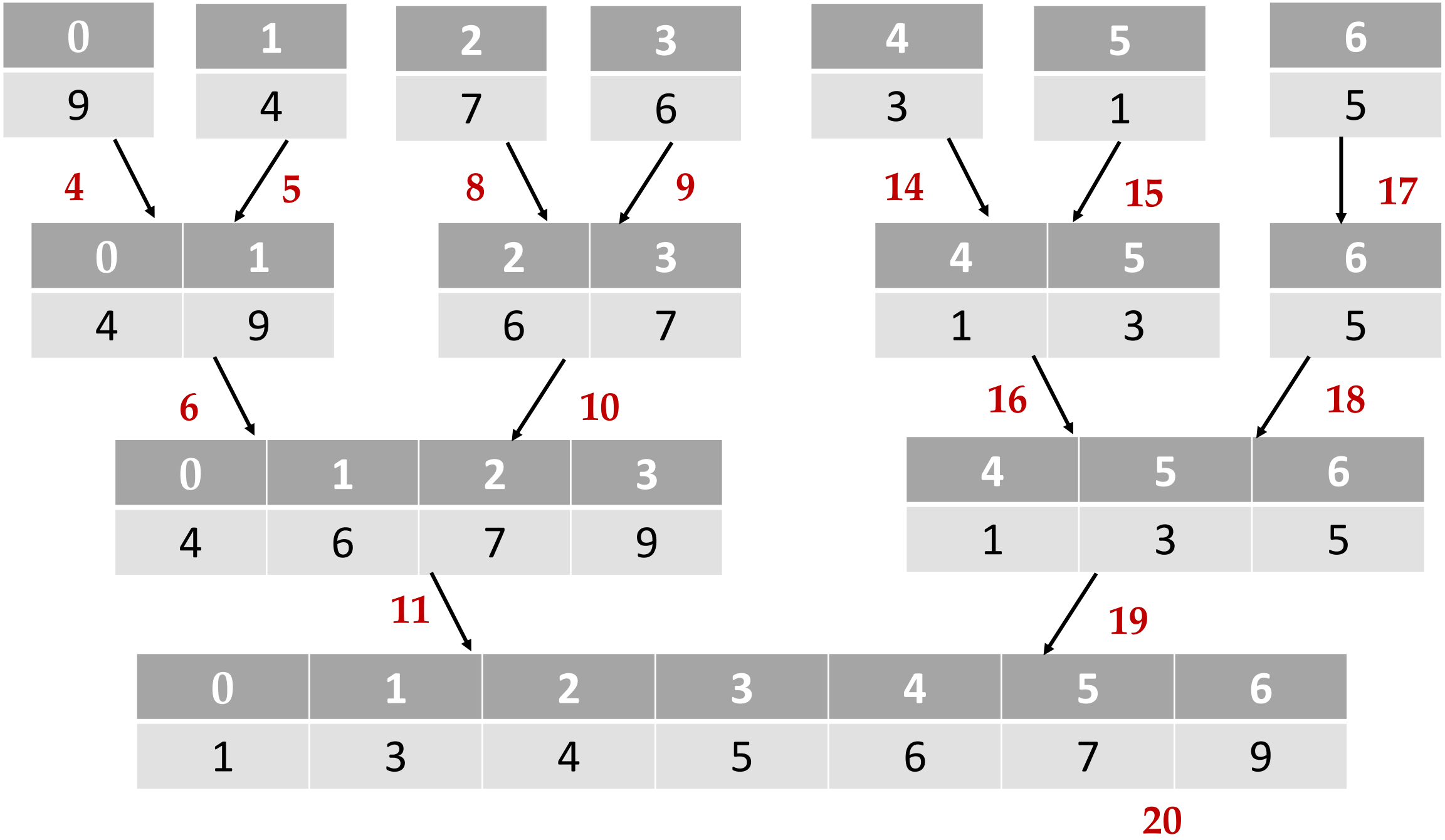
2
7

3
6

4
3

5
1

6
5



Performance – Merge Sort

- Worst case complexity : $O(n \log n)$
- Best case complexity : $O(n \log n)$
- Average case complexity : $O(n \log n)$
- Worst case space complexity: $O(n)$ auxiliary